Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	5 Marks

This document evaluates the quality of the codebase in terms of structure, readability and reusability. Update the "Followed" column after the code is written.

Planned Project Structure

```
Auto_Ticket_Classification_Using_Flow_Diagram/
  app.py                (Flask routes)
  config.py             (settings, thresholds, paths)
  requirements.txt
  .env.example
  data/                 (training CSV files)
  models/               (model.pkl, vectorizer.pkl)
  src/preprocess.py     (text cleaning)
  src/train_model.py    (training and evaluation)
  src/predict.py        (load model and predict)
  src/flow_diagram.py   (Graphviz diagram generator)
  src/routing.py        (category to team mapping)
  src/db.py             (SQLite helpers)
  templates/            (HTML pages)
  static/               (CSS, JS, generated diagrams)
  tests/                (pytest files)
```

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout (4 spaces, PEP 8)	[Update]	
2	Proper File Structure	Files and folders are logically organised as shown above	[Update]	
3	Meaningful Variable Names	Variables reflect their purpose clearly	[Update]	
4	Function / Method Names	Functions are descriptively named (e.g. clean_text, predict_category)	[Update]	
5	Code Comments	Docstrings and inline comments explain logic	[Update]	
6	Modular Design	Code is split into reusable functions/modules	[Update]	
7	No Redundant Code	Duplicate or unused code is removed	[Update]	
8	Error Handling	Exceptions such as missing model or invalid input are handled gracefully	[Update]	

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	clean_text() in preprocess.py	Python, NLTK	Training script and live prediction	High
2	predict_category() in predict.py	Python, scikit-learn	Web route, API tests, batch prediction	High
3	build_flow_diagram() in flow_diagram.py	Python, Graphviz	Result page, dashboard, reports	High
4	route_ticket() in routing.py	Python	Web route and feedback module	Medium
5	DB helper functions in db.py	Python, SQLite	All routes that read or write tickets	High